TG

Teacher Quick Start

Introduction to Matter

UNIT PURPOSE

Students build a coherent model of matter by connecting macroscopic observations with particle representations, composition, physical and chemical properties, change evidence, and property-based separation methods.

INSTRUCTIONAL RHYTHM

Engage -> Teach -> Model -> Guided Practice -> Independent Practice -> Exit Ticket -> Mixed Review -> Unit Test

SUGGESTED PACING

1. Lesson 1: Matter at Two Scales. Use the particle visual to separate observable evidence from model assumptions.
2. Lesson 2: Classify Matter by Composition. Require evidence for every label.
3. Lesson 3: Properties Describe Matter. Contrast property with change and intensive with extensive.
4. Lesson 4: Physical or Chemical Change. Treat common clues as evidence, not automatic proof.
5. Lesson 5: Separate Mixtures by Properties. Link each method to the property difference it exploits.
6. Cumulative mixed review, feedback, and targeted reteaching.
7. Unit Test A. Reserve Form B for a retake, alternate class, or test security.

PACKAGE MAP

Projectable slides teach the lesson. Student pages provide guided notes, modeled thinking, supported practice, independent practice, and exit tickets. Tests A and B follow all instruction and review. The teacher file contains every answer and rationale.

TG

Instructional Decisions

Support, monitor, and respond

BEFORE TEACHING	Preview the particle-model conventions and the documented instructional visuals. Decide whether students will work individually or discuss the engage prompts in pairs. Emphasize that a model is useful evidence but is not a literal, to-scale photograph.
FORMATIVE LOOK-FORS	Students distinguish uniform from pure, property from change, physical clue from chemical proof, and a separation method from the property difference that makes it work.
SUPPORT	Keep the composition decision path and method-property bank visible. Let students annotate particle types and regions. Provide sentence frames that require claim, evidence, and reasoning.
EXTEND	Ask students to propose an additional test that could challenge a classification, compare two viable separation sequences, or identify a limitation in a particle representation.
ASSESSMENT USE	Use Form A after instruction and mixed review. Reserve Form B for retakes, an alternate class period, or test security. Do not use either test as the first exposure to a skill.
SCOPE	Supports the NGSS practices of Developing and Using Models and Analyzing and Interpreting Data; no formal performance-expectation alignment is claimed.

1

Matter at Two Scales

TEACHER KEY

Practice answers and repair moves

L1-1. Classify air in a sealed syringe, visible light, and sound as matter or not matter. Defend each decision.

ANSWER: Air is matter because it has mass and occupies space. Visible light and sound are forms of energy/waves, not samples of matter in this classroom classification.

L1-2. In the particle visual, what evidence distinguishes a homogeneous mixture from a heterogeneous mixture?

ANSWER: Both have more than one particle type. The homogeneous mixture is uniformly distributed; the heterogeneous mixture has nonuniform regions or clustering.

L1-3. State one useful feature and one limitation of the particle visual.

ANSWER: Useful: it makes composition and distribution visible. Limitation: particle sizes, colors, spacing, and two-dimensional positions are simplified and not to scale.

L1-4. A sample expands to fill its container. Explain why this observation alone does not prove it is not matter.

ANSWER: A gas can expand to fill a container and is still matter because it has mass and occupies space.

L1-5. What additional evidence is needed to decide whether identical two-atom particles represent an element or a compound?

ANSWER: Determine whether the two atoms in each particle are the same element or different elements.

MISCONCEPTION	Only solids and liquids are matter because gases cannot always be seen.
REPAIR	**Use a sealed syringe or inflated balloon as evidence that gas occupies space and contributes mass.**

TEACHER KEY

2 Classify Matter by Composition

Practice answers and repair moves

L2-1. Classify a clean copper wire as element, compound, homogeneous mixture, or heterogeneous mixture. Explain.

ANSWER: Element. A clean copper sample consists of one chemical element, Cu.

L2-2. Classify distilled water and explain why it is not an element.

ANSWER: Compound. Pure water has hydrogen and oxygen chemically combined in a fixed composition as H2O.

L2-3. Classify clean, well-mixed air at one location. Explain why uniform does not mean pure.

ANSWER: Homogeneous mixture. It is uniform at the sampled scale but contains multiple substances, including nitrogen, oxygen, argon, and others.

L2-4. Classify granite with visibly different mineral regions.

ANSWER: Heterogeneous mixture because composition is not uniform across the visible sample.

L2-5. A model has two particle types evenly intermingled. Give the most specific label and two pieces of evidence.

ANSWER: Homogeneous mixture: more than one particle type is present, and the types are uniformly distributed.

MISCONCEPTION	A sample that looks uniform must be a pure substance.
REPAIR	**Ask whether a physical separation or particle model reveals more than one component before using the word pure.**

3 Properties Describe Matter

TEACHER KEY

Practice answers and repair moves

L3-1. Classify melting point as physical or chemical and intensive or extensive.

ANSWER: Physical and intensive under the same pressure and purity conditions.

L3-2. Classify volume as physical or chemical and intensive or extensive.

ANSWER: Physical and extensive because measuring volume does not require a new substance and volume depends on amount.

L3-3. Classify flammability and explain why it is not a physical property.

ANSWER: Chemical property. It describes the ability to react by burning and form new substances.

L3-4. A lab card says 'Copper conducts electric current.' Is this a property or a change? Classify it.

ANSWER: It is a physical property; conductivity can be observed without changing copper into a new substance.

L3-5. Why is density often more useful for identification than volume?

ANSWER: Density is intensive and can remain characteristic under stated conditions, while volume changes with sample amount.

MISCONCEPTION	The word chemical in a material's name makes every listed characteristic a chemical property.
REPAIR	**Use the identity-change test: could the characteristic be measured while the substance remains the same substance?**

4 Physical or Chemical Change?

TEACHER KEY

Practice answers and repair moves

L4-1. Ice melts in a sealed bag. Classify the change and identify what remains the same.

ANSWER: Physical change. The substance remains H2O; only its physical state changes.

L4-2. Candle wax burns in oxygen and produces different substances. Classify the change and cite decisive evidence.

ANSWER: Chemical change because atoms are rearranged into new substances such as carbon dioxide and water vapor.

L4-3. A tablet is crushed into powder without reacting. Classify the change.

ANSWER: Physical change; size and surface area change, but no new substance is required.

L4-4. Explain why bubbles support but do not by themselves prove a chemical-change claim.

ANSWER: Bubbles may be a gas product, but boiling or release of dissolved gas can also make bubbles; context and additional evidence are needed.

L4-5. A solid is strongly heated in an open container and a gas escapes. Why can the measured solid mass decrease without violating conservation?

ANSWER: Some product matter leaves as gas. Total matter is conserved when all products are included, but the open container does not retain every product.

MISCONCEPTION	Any color change, temperature change, or bubble formation automatically proves a chemical reaction.
REPAIR	**Require students to name the proposed new substance or rule out a physical alternative before accepting the claim.**

5 Separate Mixtures by Properties

TEACHER KEY

Practice answers and repair moves

L5-1. Choose a method to separate insoluble sand from water and name the property difference.

ANSWER: Filtration; sand is insoluble and its particles are retained while water passes through.

L5-2. A saltwater sample must yield both water and salt. Choose a method and explain why evaporation alone is incomplete.

ANSWER: Distillation. It collects condensed water and leaves salt; evaporation alone may recover salt but does not collect the water.

L5-3. Choose a method to test whether black marker ink contains more than one dye.

ANSWER: Paper chromatography; components travel differently because of different attractions to the mobile and stationary phases.

L5-4. Choose the first method for a dry mixture of iron filings and table salt.

ANSWER: Magnetic separation; a magnet attracts iron but not salt.

L5-5. Explain why separating a mixture is usually physical even when several steps are required.

ANSWER: The methods exploit physical-property differences and recover components without requiring their chemical identities to change.

MISCONCEPTION	One separation method should remove every component from any mixture.
REPAIR	**Build a method-property table and choose one step for one property difference at a time; then reevaluate the remaining mixture.**

R

Cumulative Review Key

Answers from all lessons

TEACHER KEY

R1. Explain why air is matter even though it is usually invisible.

ANSWER: Air has mass and occupies space; visibility is not part of the definition of matter.

R2. A particle model shows one repeated particle made of two different atom types. Classify the sample.

ANSWER: A pure compound: one particle type is present, and each particle contains different elements chemically combined.

R3. Classify salad dressing with visible oil and water layers.

ANSWER: Heterogeneous mixture because composition is nonuniform and distinct phases are visible.

R4. Classify electrical conductivity and flammability as physical or chemical properties.

ANSWER: Conductivity is physical; flammability is chemical.

R5. Explain why mass is extensive but density is intensive.

ANSWER: Mass depends on sample amount; density can remain characteristic as amount changes under the same conditions.

R

Cumulative Review Key

Answers from all lessons

R6. Classify boiling water and burning paper as physical or chemical changes.

ANSWER: Boiling water is physical; burning paper is chemical because new substances form.

R7. Bubbles form when a liquid is warmed. What additional evidence is needed before claiming a chemical change?

ANSWER: Rule out boiling or released dissolved gas and seek evidence of a new substance, such as a product with a new composition or properties.

R8. Choose a method for oil and water layers and name the property evidence.

ANSWER: Decantation or a separatory funnel; immiscibility creates separate phases, and density helps determine their layering.

R9. A distillation starts with 120.0 g and recovers 92.0 g liquid plus 27.6 g solid. Calculate percent recovery.

ANSWER: (92.0 + 27.6) / 120.0 x 100 = 99.7%.

R10. Why can a uniform appearance support but not prove a homogeneous-mixture classification?

ANSWER: Appearance shows uniformity only at the observed scale; purity and dissolved components require additional evidence.

A

Unit Test A - Rationale Key

TEACHER KEY

Ten items per page

1. C - Air in a sealed syringe

Air has mass and occupies space, so it is matter.

2. D - Homogeneous mixture

More than one particle type indicates a mixture, and uniform distribution makes it homogeneous.

3. A - Particle size and spacing are not shown to scale.

Classroom particle diagrams simplify size, spacing, color, and dimensionality.

4. D - A physical process recovers two components.

Physical recovery of components supports the presence of more than one substance.

5. A - Element

Clean copper consists of one chemical element, Cu.

6. B - Compound

Water contains hydrogen and oxygen chemically combined in a fixed composition.

7. C - It contains multiple substances.

Air can be uniform while containing nitrogen, oxygen, argon, and other substances.

8. D - heterogeneous mixture

Distinct mineral regions show nonuniform composition.

9. B - Density

Density can be measured without changing identity and does not depend on sample amount under the same conditions.

10. D - Flammability

Flammability describes the ability to react by burning and form new substances.

A

Unit Test A - Rationale Key

Ten items per page

TEACHER KEY

11. A - Mass

Mass is extensive and changes with the amount of sample.

12. C - Ice melts.

Melting changes physical state while the substance remains H_2O.

13. D - New substances with different compositions form.

Formation of new substances is the defining criterion for a chemical change.

14. A - Boiling can also produce bubbles.

A phase change or release of dissolved gas can produce bubbles without a chemical reaction.

15. B - Filtration

Filtration retains insoluble particles while the liquid passes through.

16. D - Distillation

Distillation collects condensed water while salt remains.

17. A - Chromatography

Chromatography separates soluble components by different attractions and travel rates.

18. C - Use a magnet.

Magnetism selectively removes iron before additional steps are needed.

19. B - 99.7%

Recovered mass is 119.6 g; 119.6 / 120.0 x 100 = 99.7%.

20. C - A gaseous product may leave the container.

An open system may not retain gaseous products; total matter is conserved when all products are included.

B

Unit Test B - Rationale Key

TEACHER KEY

Ten items per page

1. D - It is visible.

Visibility is not part of the definition; invisible gases can be matter.

2. A - Compound

One repeated particle type supports purity; different elements bonded in each particle support a compound.

3. C - Composition and distribution

Symbols and arrangement provide model evidence about particle types and distribution.

4. B - homogeneous mixture

The clear sample is uniform, while physical recovery of components shows it is a mixture.

5. A - Distilled H2O

Pure water contains hydrogen and oxygen chemically combined in a fixed ratio.

6. D - Well-mixed air

Well-mixed air contains multiple substances distributed uniformly at the sampled scale.

7. B - Oil and water layers

Visible immiscible layers show nonuniform composition.

8. C - A physical method separates two components.

Recovering components physically supports a mixture rather than a pure substance.

9. A - Density and conductivity

Density and conductivity can be measured without changing chemical identity.

10. C - Volume

Volume depends on the amount of sample present.

Unit Test B - Rationale Key

TEACHER KEY

Ten items per page

11. A - It describes potential chemical behavior.

A property can describe a substance's ability to undergo a change without the change occurring during the statement.

12. D - Wax burns and forms new substances.

Burning rearranges atoms and forms products with new compositions.

13. B - Bubbles appear.

Bubbles can also result from boiling or release of dissolved gas.

14. A - Its size changes without requiring a new substance.

Crushing changes size and surface area, not chemical identity.

15. C - Decantation or a separatory funnel

Separate phases can be drained or poured apart; density helps determine layering.

16. D - Dissolved particles pass with the liquid.

A filter retains insoluble particles but does not ordinarily remove dissolved solute.

17. B - Volatility or boiling behavior

Components vaporize and condense differently because their volatility or boiling behavior differs.

18. A - Use a magnet, then evaluate the sand.

Magnetism selectively removes iron from a dry nonmagnetic solid.

19. C - It is a homogeneous mixture with dissolved material.

No residue is expected for dissolved solute, while distillation physically recovers components.

20. B - Gas products count even if they leave an open container.

All products, including gases, must be included when accounting for matter.

